

Predictive Multi-Warehouse Inventory Optimization & Stockout Risk Control Tower

A Business Analytics Framework for Stockout Prediction, Inventory Rebalancing and Replenishment Decision Support

Introduction

Inventory management in a multi-warehouse supply chain is challenging because the same product may be overstocked in one warehouse while facing shortage risk in another. These imbalances can increase holding costs, create unnecessary replenishment requirements and increase the risk of stockouts. Supplier lead times, changing demand and warehouse-level inventory differences further complicate inventory planning.

This project, **Predictive Multi-Warehouse Inventory Optimization & Stockout Risk Control Tower**, develops an end-to-end business analytics framework for identifying stockout risk, predicting vulnerable inventory positions and recommending inventory rebalancing or replenishment actions.

The analysis uses SKU-, warehouse-, supplier-, demand- and inventory-level data. The main target variable is **Model_Stockout_Flag**, where 0 = No Stockout and 1 = Stockout, representing whether a SKU–warehouse observation experiences stockout risk within the supplier lead-time horizon. The final modeling dataset contains **89,254 observations**, including **3,242 stockout cases (3.6%)**, making stockout prediction an imbalanced classification problem.

Project Objectives

- Identify the major operational factors associated with stockout risk.
- Examine whether stockout rates differ across warehouses and suppliers.
- Develop and compare predictive models for identifying stockout-risk cases.
- Identify low-stock positions that can be supported using excess inventory from other warehouses.

- Recommend **Transfer, Reorder, Transfer + Reorder, or No Action** for inventory positions.
- Evaluate supply-chain vulnerability under demand, inventory and supplier lead-time disruptions.
- Translate inventory rebalancing into operational and financial impact.
- Present the results through an interactive decision-support dashboard.

Analytics Tools Used

Tool	Analytics Role	What is done in the project
Jamovi	Diagnostic Analytics	Descriptive statistics, stockout frequency analysis, warehouse/supplier chi-square tests and binomial logistic regression to identify stockout drivers
Orange	Predictive Analytics	Compare Logistic Regression, Random Forest and Decision Tree using cross-validation, confusion matrix and ROC analysis
Python	Prescriptive & Scenario Analytics	Generate warehouse transfer recommendations, replenishment decisions, priority levels, stress tests, network-health measures and financial impact
Streamlit	Decision Support	Convert the analytical outputs into an interactive Supply Chain Control Tower with KPIs, stress testing and SKU–warehouse decision lookup

Together, the four tools create a progression from **Diagnostic** → **Predictive** → **Prescriptive** → **Decision-Support Analytics**

Stage 1: Diagnostic Analytics Using Jamovi

Purpose of using Jamovi:

Jamovi is used to understand the structure of the dataset and identify the factors associated with stockout risk before predictive modeling. The analysis includes descriptive statistics, stockout-frequency analysis, chi-square tests and binomial logistic regression.

Questions addressed:

- How frequently do stockouts occur?
- Do stockout rates differ by warehouse or supplier?
- Which inventory, demand and lead-time factors significantly influence stockout risk?

Dataset Preparation

- **Unit of analysis:** SKU–Warehouse observation
- **Modeling observations:** 89,254
- **Target:** `Model_Stockout_Flag`
- **Target classes:** 0 = No Stockout, 1 = Stockout
- **Stockout observations:** 3,242 (3.6%)
- **Non-stockout observations:** 86,012 (96.4%)
- Variables covering inventory, supplier lead time, demand, promotion, warehouse and supplier characteristics were used.
- Future information used to construct the target was excluded from the predictive model to avoid **data leakage**.

Variable	Description	Measure type
Date	Date of the inventory observation	ID / Nominal
SKU_ID	Unique identifier for each product/SKU	Nominal
Warehouse_ID	Warehouse in which the SKU is stored	Nominal
Supplier_ID	Supplier associated with the SKU	Nominal
Region	Geographic region of the observation	Nominal
Inventory_Level	Number of units currently available in inventory	Continuous
Supplier_Lead_Time_Days	Number of days required by the supplier to replenish inventory	Continuous
Reorder_Point	Inventory level at which replenishment should be triggered	Continuous
Demand_Forecast	Forecasted demand for the SKU	Continuous
Promotion_Flag	Indicates whether a promotion is active: 0 = No, 1 = Yes	Nominal
Rolling_7D_Avg_Demand	Average recent demand over the current and previous 6 observations for the same SKU–warehouse combination	Continuous
Rolling_14D_Avg_Demand	Average recent demand over the current and previous 13 observations	Continuous
Inventory_Coverage_Days	Estimated number of days current inventory can support recent demand	Continuous
Lead_Time_Demand_Forecast	Expected demand during the supplier lead-time period	Continuous
Below_Reorder_Point	Indicates whether inventory is below its reorder point: 0 = No, 1 = Yes	Nominal
Forecast_Error	Difference between actual demand and forecast demand	Continuous
Model_Stockout_Lag	Indicates whether inventory is expected to stock out within the supplier lead-time horizon: 0 = No, 1 = Yes	Nominal

Descriptives

Descriptives

	Inventory_ Level	Supplier_Lead_Ti me_Days	Reorder_ Point	Demand_Fo recast	Rolling_7D_Avg_ Demand	Inventory_Covera ge_Days
Mean	471.462	7.941	300.05 2	20.097	20.087	28.148
Medi an	461.000	8.000	299.00 0	19.980	20.286	23.599
Stand ard devia tion	133.603	3.908	54.869	9.564	7.615	15.801
Mini mum	168	2	201	0.000	3.714	4.992
Maxi mum	990	14	398	61.420	41.000	191.852

The descriptive statistics show that the supply chain operates with an average inventory level of about **471 units**, while the average reorder point is about **300 units**. This suggests that, overall, inventory is usually maintained above the reorder threshold. However, the wide range in inventory levels, from **168 to 990 units**, indicates that some SKU–warehouse combinations may hold substantially more or less stock than others.

Supplier lead time averages about **8 days**, but ranges from **2 to 14 days**. This variation is operationally important because items supplied by longer-lead-time suppliers may be more exposed to stockout risk if demand rises unexpectedly or replenishment is delayed.

The average demand forecast is **20.10 units**, while the rolling 7-day average demand is **20.09 units**. At an overall level, this suggests that forecasted demand is quite close to recent actual demand. However, this does not mean the forecast is accurate for every SKU or warehouse; it only shows that the averages are similar across the full dataset.

Inventory coverage averages **28.15 days**, with a median of **23.60 days**. Since the mean is noticeably higher than the median and the maximum reaches almost **192 days**, some observations appear to hold very large amounts of inventory relative to their demand. This suggests the presence of possible **excess inventory** in parts of the warehouse network.

At the same time, the minimum inventory coverage is only about **5 days**, which means some SKU–warehouse combinations have far less protection against future demand. Therefore, the dataset appears to contain both **overstocked and relatively**

vulnerable inventory positions, which supports the need for warehouse rebalancing and stockout-risk analysis.

The network does not have a uniform inventory position. Some SKU–warehouse combinations appear overstocked, while others have much lower coverage, creating an opportunity for inventory rebalancing and targeted replenishment.

Stockout Frequency Distribution

Frequencies

Frequencies of Model_Stockout_Flag

Model_Stockout_Flag	Counts	% of Total	Cumulative %
0	86012	96.4%	96.4%
1	3242	3.6%	100.0%

Purpose of the analysis: The frequency distribution of Model_Stockout_Flag was examined to understand how frequently stockout events occur in the dataset and to assess whether the target variable is balanced before developing predictive models.

The stockout target is highly imbalanced. Out of **89,254 observations**, **86,012 (96.4%)** were classified as non-stockout cases, while **3,242 (3.6%)** were classified as stockout cases. This indicates that stockout events are relatively rare compared with normal inventory observations.

This imbalance is important for the later predictive modeling stage because overall accuracy alone may be misleading. For example, a model that predicts “no stockout” for almost every observation could still achieve a very high accuracy. Therefore, the later model comparison will focus on measures such as **recall, precision, F1-score, and ROC-AUC** in addition to accuracy.

Warehouse-Level Stockout Analysis

Purpose of the analysis: The contingency table was used to examine whether the proportion of stockout cases differs across warehouses and whether warehouse location is significantly associated with stockout status.

Contingency Tables

Contingency Tables

Warehouse_ID		Model_Stockout_Flag		Total
		0	1	
WH_1	Observed	17185	665	17850
	% within row	96.3%	3.7%	100.0%
WH_2	Observed	17274	553	17827
	% within row	96.9%	3.1%	100.0%
WH_3	Observed	17055	784	17839
	% within row	95.6%	4.4%	100.0%
WH_4	Observed	17031	795	17826
	% within row	95.5%	4.5%	100.0%
WH_5	Observed	17467	445	17912
	% within row	97.5%	2.5%	100.0%
Total	Observed	86012	3242	89254
	% within row	96.4%	3.6%	100.0%

χ^2 Tests

	Value	df	p
χ^2	146.699	4	<.0001
N	89254		

Stockout rates vary across the five warehouses. WH_4 recorded the highest stockout rate at **4.5%**, followed by WH_3 at **4.4%**, while WH_5 had the lowest rate at **2.5%**. The chi-square test of independence was statistically significant, $\chi^2(4) = 146.699$, $p < .0001$, indicating that stockout status is significantly associated with warehouse location.

However, this does not mean that the warehouse itself causes stockouts. Differences in inventory levels, supplier lead times, and demand conditions may also contribute to the observed variation. These factors will be examined together in the logistic regression analysis.

Supplier-Level Stockout Analysis

Purpose of the analysis: The supplier contingency table was used to examine whether stockout rates differ across suppliers and whether supplier identity is significantly associated with stockout status.

Contingency Tables

Contingency Tables

Supplier_ID		Model_Stockout_Flag		Total
		0	1	
SUP_1	% within row	95.6%	4.4%	100.0%
SUP_10	% within row	97.8%	2.2%	100.0%
SUP_2	% within row	94.8%	5.2%	100.0%
SUP_3	% within row	94.8%	5.2%	100.0%
SUP_4	% within row	96.8%	3.2%	100.0%
SUP_5	% within row	96.8%	3.2%	100.0%
SUP_6	% within row	97.8%	2.2%	100.0%
SUP_7	% within row	97.3%	2.7%	100.0%
SUP_8	% within row	95.2%	4.8%	100.0%
SUP_9	% within row	97.0%	3.0%	100.0%
Total	% within row	96.4%	3.6%	100.0%

χ^2 Tests

	Value	df	p
χ^2	334.888	9	<.0001
N	89254		

Stockout rates varied across suppliers. SUP_2 and SUP_3 recorded the highest stockout rates at **5.2%**, followed by SUP_8 at **4.8%** and SUP_1 at **4.4%**. The lowest stockout rates were observed for SUP_6 and SUP_10, both at **2.2%**.

The chi-square test of independence was statistically significant, $\chi^2(9) = 334.888$, $p < .0001$, indicating that stockout status is significantly associated with supplier identity.

This suggests that some suppliers are linked with greater stockout exposure than others. However, supplier identity alone does not explain the reason for this difference. Variables such as supplier lead time, inventory level, demand, and promotion activity may contribute to the observed supplier-level variation and will be examined in the logistic regression analysis.

Binomial Logistic Regression Analysis

Purpose of the analysis: A binomial logistic regression was conducted to identify which operational factors significantly influence the probability of a stockout while controlling for the effects of the other variables in the model. The dependent variable was Model_Stockout_Flag (0 = no stockout, 1 = stockout). Inventory level,

supplier lead time, demand forecast, recent 7-day average demand, promotion status, warehouse, and supplier were included as predictors.

Model Fit Measures

Model	Deviance	R^2_{McF}	Overall Model Test		
			χ^2	df	p
1	5042.790	0.819	22818.376	18	<.0001

Note. Models estimated using sample size of N=89254

The overall logistic regression model was statistically significant, $\chi^2(18) = 22,818.376$, $p < .0001$, indicating that the predictors collectively provide substantial information for distinguishing between stockout and non-stockout observations. The McFadden pseudo- R^2 value of **0.819** indicates very strong explanatory performance for this dataset.

The **Omnibus Likelihood Ratio Tests** are very important because they tell us whether each predictor contributes significantly after controlling for all the others.

Omnibus Likelihood Ratio Tests

Predictor	χ^2	df	p
Inventory_Level	11907.866	1	<.0001
Supplier_Lead_Time_Days	10024.462	1	<.0001
Demand_Forecast	6.566	1	.0104
Rolling_7D_Avg_Demand	3529.977	1	<.0001
Promotion_Flag	42.493	1	<.0001
Warehouse_ID	7.111	4	.1301
Supplier_ID	15.162	9	.0866

Earlier, our contingency tables showed warehouse and supplier were significantly associated with stockout status.

But after controlling for **inventory, lead time and demand**, warehouse and supplier are **no longer statistically significant overall**.

It suggests that the apparent differences between warehouses and suppliers may largely be explained by underlying operating conditions such as inventory availability, supplier lead time and demand rather than simply by warehouse or supplier identity itself.

That's exactly why logistic regression adds value beyond contingency tables.

Model Coefficients - Model_Stockout_Flag

Predictor	Estimate	95% Confidence Interval		SE	Z	p	Odds ratio
		Lower	Upper				
Intercept	-17.625	-18.535	-16.715	0.464	-37.959	<.0001	0.000
Inventory_Level	-0.056	-0.058	-0.053	0.001	-47.035	<.0001	0.946
Supplier_Lead_Time_Days	1.612	1.541	1.682	0.036	44.954	<.0001	5.011
Demand_Forecast	0.016	0.004	0.029	0.006	2.561	.0104	1.017
Rolling_7D_Avg_Demand	0.575	0.547	0.603	0.014	39.831	<.0001	1.777
Promotion_Flag:							
1 – 0	-0.817	-1.067	-0.567	0.128	-6.405	<.0001	0.442
Warehouse_ID:							
WH_2 – WH_1	-0.252	-0.494	-0.010	0.123	-2.039	.0414	0.777
WH_3 – WH_1	-0.209	-0.430	0.012	0.113	-1.855	.0636	0.811
WH_4 – WH_1	-0.187	-0.414	0.039	0.116	-1.619	.1055	0.829
WH_5 – WH_1	-0.291	-0.540	-0.041	0.127	-2.284	.0224	0.748
Supplier_ID:							
SUP_10 – SUP_1	0.245	-0.093	0.583	0.173	1.422	.1550	1.278
SUP_2 – SUP_1	-0.114	-0.410	0.182	0.151	-0.754	.4506	0.892
SUP_3 – SUP_1	0.074	-0.222	0.370	0.151	0.491	.6233	1.077
SUP_4 – SUP_1	0.093	-0.217	0.404	0.158	0.589	.5562	1.098
SUP_5 – SUP_1	0.293	-0.042	0.628	0.171	1.714	.0866	1.340
SUP_6 – SUP_1	0.068	-0.294	0.430	0.185	0.368	.7125	1.070
SUP_7 – SUP_1	0.221	-0.091	0.532	0.159	1.387	.1655	1.247
SUP_8 – SUP_1	-0.046	-0.333	0.241	0.146	-0.315	.7531	0.955
SUP_9 – SUP_1	0.377	0.024	0.730	0.180	2.091	.0365	1.457

Note. Estimates represent the log odds of "Model_Stockout_Flag = 1" vs. "Model_Stockout_Flag = 0"

The regression results show that stockout risk is driven primarily by operational conditions rather than by warehouse or supplier identity alone. **Higher inventory levels significantly reduce stockout risk**, while **longer supplier lead times and stronger recent demand significantly increase it**. Supplier lead time is especially important, with each additional day associated with roughly **5 times higher odds of stockout**, making replenishment speed a major risk factor. Recent 7-day demand is also a strong predictor, indicating that sudden

increases in actual demand can create stockout pressure even when forecast demand is available.

After controlling for these factors, most warehouse and supplier differences are no longer significant. This suggests that the earlier differences seen across warehouses and suppliers are largely explained by underlying inventory, demand, and lead-time conditions. For management, this means stockout prevention should focus more on **maintaining adequate inventory coverage, responding quickly to rising demand, and reducing long supplier lead times** rather than treating certain warehouses or suppliers as inherently risky.

Classification Table – ...

Observed	Predicted		% Correct
	0	1	
0	8556 6	446	99.481
1	642	2600	80.197

Note. The cut-off value is set to 0.5

Predictive Measures

Accuracy	Specificity	Sensitivity	AUC
0.988	0.995	0.802	0.99 6

Note. The cut-off value is set to 0.5

The logistic regression model performs very strongly overall, with an accuracy of **98.8%** and an AUC of **0.996**, indicating excellent ability to distinguish between stockout and non-stockout cases. More importantly, the model correctly identifies about **80.2% of actual stockout cases** (sensitivity), while correctly identifying **99.5% of non-stockout cases**(specificity). This is important because the dataset is highly imbalanced; therefore, the strong sensitivity shows that the model is not simply achieving high accuracy by predicting “no stockout” for most observations. The model is able to detect a substantial majority of stockout-risk cases and can therefore be useful for early inventory intervention.

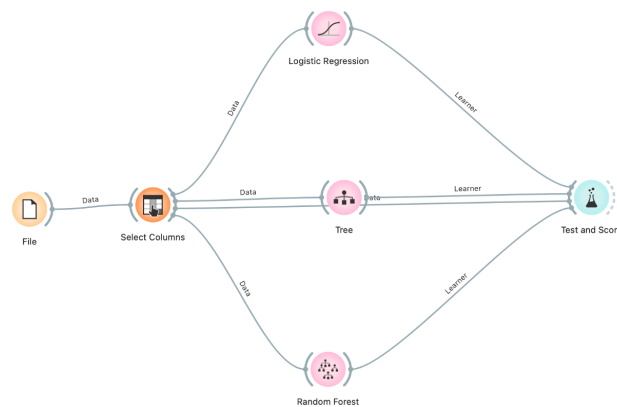
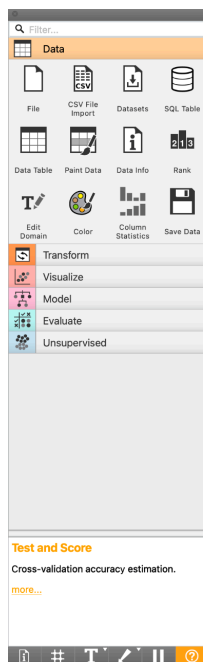
Stage 2: Predictive Analytics Using Orange

Purpose of using Orange:

Orange is used to evaluate and compare alternative classification models for predicting stockout-risk cases. Because only **3.6%** of observations are stockouts, model evaluation focuses on **ROC-AUC, Recall, Precision, F1-score and MCC**, rather than accuracy alone.

Models compared:

- Logistic Regression
- Random Forest
- Decision Tree



Orange was used to compare multiple classification models and identify which model most effectively predicts stockout cases, particularly because stockouts represent only a small proportion of the dataset.

Cross validation

Number of folds: 10

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target 1

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.996	0.988	0.826	0.853	0.800	0.820
Random Forest	0.988	0.986	0.802	0.857	0.754	0.797
Tree	0.500	0.964	0.000	0.000	0.000	0.000

Compare models by: Area under ROC curve

Negligible diff.: 0.1

	Logistic Regression	Random Forest	Tree
Logistic Regression		1.000	1.000
Random Forest	0.000		1.000
Tree	0.000	0.000	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Logistic Regression provided the strongest overall performance, achieving the highest **AUC (0.996)**, **F1-score (0.826)**, **recall (0.800)** and **MCC (0.820)**. This means it was able to identify approximately **80% of actual stockout cases** while maintaining strong overall classification performance. Random Forest performed slightly worse, particularly in recall, identifying about **75.4%** of stockout cases. The Decision Tree performed poorly, with an AUC of **0.500** and zero recall for stockouts, despite showing 96.4% accuracy. This demonstrates why accuracy alone is misleading for an imbalanced dataset. Based on these results, **Logistic Regression is retained as the preferred predictive model for the project.**

Learners

Logistic Regression

Random Forest

Tree

Clicking on cells or in headers outputs the corresponding data instances

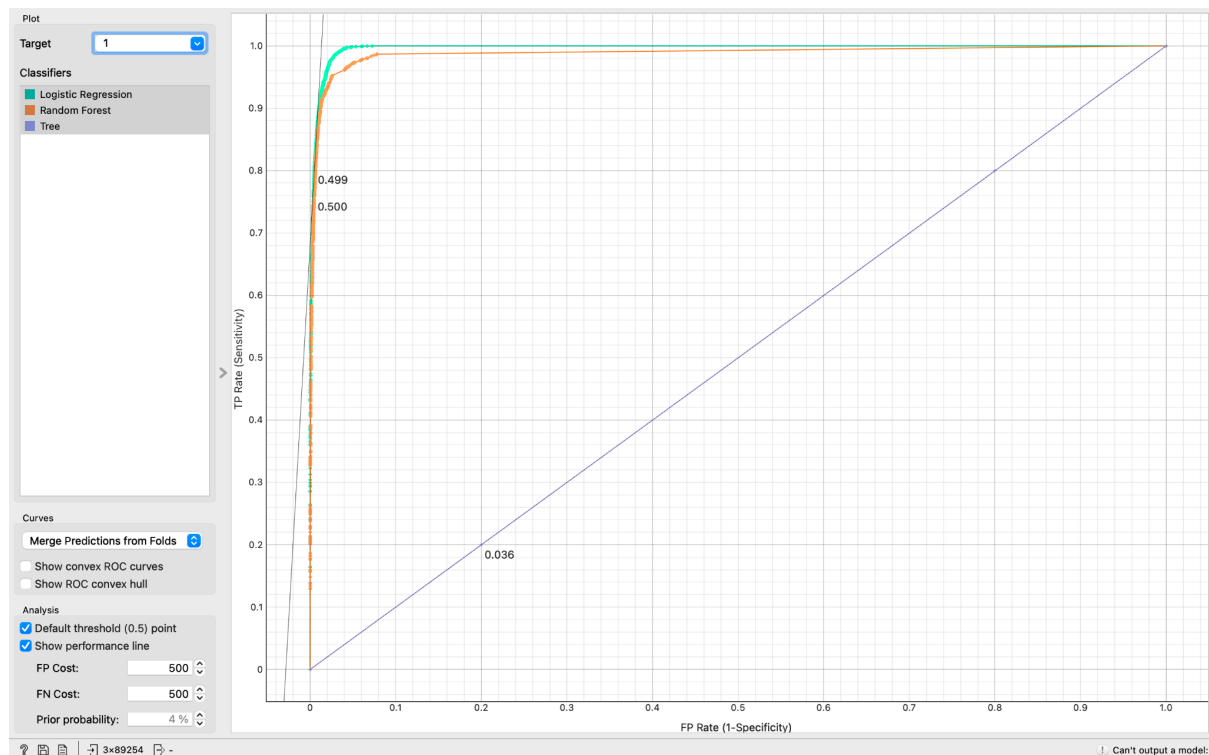
Ok, got it

Show: Number of instances

		Predicted		Σ
		0	1	
Actual	0	85564	448	86012
	1	647	2595	3242
	Σ	86211	3043	89254

Purpose of analysis: The confusion matrix was used to examine how well the selected Logistic Regression model distinguishes between stockout and non-stockout cases.

The model correctly classified **85,564 non-stockout observations** and **2,595 stockout observations**. It missed **647 actual stockouts** and incorrectly flagged **448 non-stockout cases** as stockouts. This shows that the model is effective at detecting most stockout cases while keeping false alarms relatively low. Since the business objective is to identify potential stockouts early, correctly identifying **2,595 of 3,242 actual stockouts** is a meaningful result.



ROC analysis was used to compare how well the three classification models distinguish stockout cases from non-stockout cases across different probability thresholds.

Logistic Regression shows the strongest discrimination performance, with its ROC curve closest to the top-left corner and an AUC of **0.996**. Random Forest also performs strongly with an AUC of **0.988**, but is slightly weaker. The Decision Tree follows the diagonal line with an AUC of **0.500**, indicating performance close to random classification. Therefore, Logistic Regression is selected as the preferred predictive model for stockout-risk identification.

Cross-Software Validation

The same Logistic Regression framework was evaluated in Jamovi and Orange using the same target variable and predictor set. The results were highly consistent across both software packages.

Measure	Jamovi	Orange
ROC-AUC	0.996	0.996
Stockout Recall / Sensitivity	0.802	0.800
Correctly identified stockouts	2,600	2,595
Overall Accuracy	0.988	0.988

The small difference in correctly classified observations occurs because Orange reports results from **10-fold cross-validation**, while the Jamovi classification table evaluates the fitted model at a 0.50 cut-off. Both packages therefore provide closely matching evidence of strong predictive performance.

Prediction on New Data

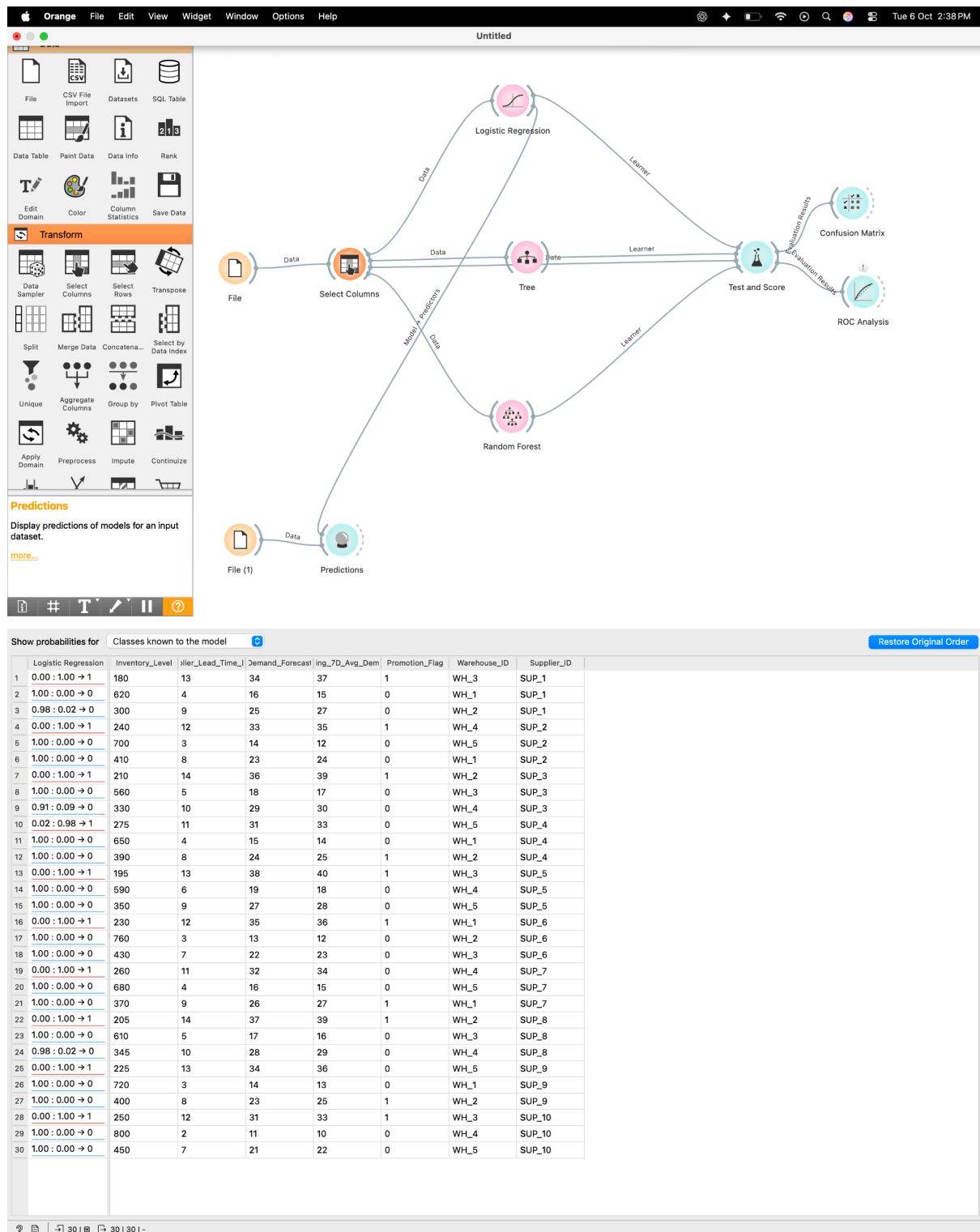
After selecting Logistic Regression as the best-performing model in Orange, the model was applied to a separate set of new inventory observations to demonstrate how it could be used in practice. These new observations contained the same predictor variables used during model development, such as inventory level, supplier lead time, demand forecast, recent demand, promotion status, warehouse and supplier, but they did not contain the actual stockout outcome.

The purpose of this step was to test how the trained model would behave when new inventory information becomes available. For each new SKU–warehouse observation, Orange generated both a predicted class and the probability of belonging to each class. A prediction of **1** represents a likely stockout, while **0** represents no stockout.

The results showed a clear pattern. Observations with lower inventory levels, longer supplier lead times and stronger recent demand were generally assigned higher probabilities of stockout. In contrast, observations with higher inventory levels, shorter replenishment times and lower demand were generally classified as non-stockout cases. This is consistent with the relationships identified earlier in the Jamovi analysis.

For example, a new observation with relatively low inventory, a long supplier lead time and high recent demand was predicted as a stockout-risk case. On the other hand, an observation with high inventory, short lead time and lower recent demand was predicted as a non-stockout case. This shows that the model is able to use previously learned relationships and apply them to new operational situations.

Since these new observations do not contain the actual future stockout outcome, this stage is used to demonstrate **practical prediction rather than model accuracy**. The actual stockout result would only become known after the future lead-time period has passed. Model performance had already been evaluated using 10-fold cross-validation, while this additional step demonstrates how the selected model can be used on new incoming business data.



Application:

This prediction step can act as an early-warning mechanism for inventory management. When new warehouse data becomes available, the model can identify SKU–warehouse combinations that appear likely to stock out before the shortage actually occurs. These predicted high-risk cases can then be passed to the Python decision engine, which can recommend whether stock should be transferred from

another warehouse, reordered from the supplier, or handled using a combination of both actions.

In this way, the predictive model supports the overall project objective of moving from simply identifying stockout risk to taking preventive inventory action before the problem occurs.

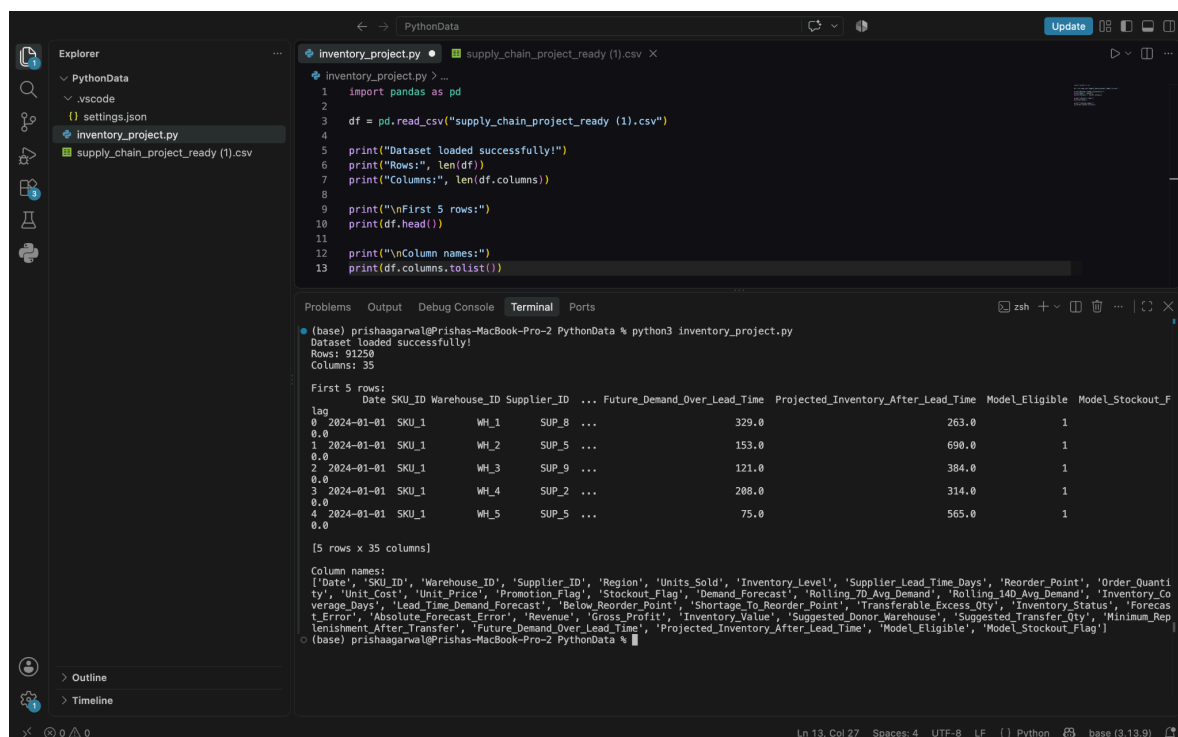
Stage 3: Prescriptive and Scenario Analytics Using Python

Purpose of using Python:

Python is used to convert stockout and inventory information into operational decisions. Rather than only identifying risk, this stage determines how the supply-chain network should respond.

Python is used to:

- classify inventory as healthy, low or excess;
- identify possible donor warehouses;
- recommend inter-warehouse transfer quantities;
- calculate remaining supplier replenishment;
- assign inventory priority levels;
- generate recommended actions;
- perform disruption stress testing;
- evaluate network health before and after rebalancing; and
- estimate the financial value of inventory covered through internal transfers.



The screenshot shows a VS Code editor with a Python script named `inventory_project.py` and its terminal output. The script reads a CSV file, prints dataset statistics, and displays the first 5 rows of data.

```
1 import pandas as pd
2
3 df = pd.read_csv("supply_chain_project_ready (1).csv")
4
5 print("Dataset loaded successfully!")
6 print("Rows:", len(df))
7 print("Columns:", len(df.columns))
8
9 print("\nFirst 5 rows:")
10 print(df.head())
11
12 print("\nColumn names:")
13 print(df.columns.tolist())
```

Terminal Output:

```
(base) prishaagarwal@Prishas-MacBook-Pro-2 PythonData % python3 inventory_project.py
Dataset loaded successfully!
Rows: 91250
Columns: 35

First 5 rows:
   lag  Date  SKU_ID  Warehouse_ID  Supplier_ID  ...  Future_Demand_Over_Lead_Time  Projected_Inventory_After_Lead_Time  Model_Eligible  Model_Stockout_F
0.0  2024-01-01  SKU_1           WH_1         SUP_8  ...                      329.0                      263.0                1
1.0  2024-01-01  SKU_1           WH_2         SUP_5  ...                      153.0                      690.0                1
2.0  2024-01-01  SKU_1           WH_3         SUP_9  ...                      121.0                      384.0                1
3.0  2024-01-01  SKU_1           WH_4         SUP_2  ...                      208.0                      314.0                1
4.0  2024-01-01  SKU_1           WH_5         SUP_5  ...                      75.0                       565.0                1

[5 rows x 35 columns]

Column names:
[Date, 'SKU_ID', 'Warehouse_ID', 'Supplier_ID', 'Region', 'Units_Sold', 'Inventory_Level', 'Supplier_Lead_Time_Days', 'Reorder_Point', 'Order_Quant
ty', 'Unit_Cost', 'Unit_Price', 'Promotion_Flag', 'Stockout_Flag', 'Demand_Forecast', 'Rolling_7D_Avg_Demand', 'Rolling_14D_Avg_Demand', 'Inventory_Co
verage_Days', 'Lead_Time_Demand_Forecast', 'Below_Reorder_Point', 'Shortage_To_Reorder_Point', 'Transferable_Excess_Qty', 'Inventory_Status', 'Forecas
t_Error', 'Absolute_Forecast_Error', 'Revenue', 'Gross_Profit', 'Inventory_Value', 'Suggested_Donor_Warehouse', 'Suggested_Transfer_Qty', 'Minimum_Rep
lenishment_After_Transfer', 'Future_Demand_Over_Lead_Time', 'Projected_Inventory_After_Lead_Time', 'Model_Eligible', 'Model_Stockout_Flag']
```

```

INVENTORY STATUS SUMMARY
-----
Inventory_Status
EXCESS    67118
HEALTHY   19345
LOW       4787
Name: count, dtype: int64

Transfer opportunities: 4754
(base) prishaagarwal@Prishas-MacBook-Pro-2 PythonData % █

SAMPLE TRANSFER RECOMMENDATIONS
-----
SKU_ID Warehouse_ID Suggested_Donor_Warehouse Suggested_Transfer_Qty Minimum_Replenishment_After_Transfer
SKU_16 WH_5 WH_1 4 0
SKU_13 WH_2 WH_4 4 0
SKU_18 WH_5 WH_3 5 0
SKU_41 WH_1 WH_3 12 0
SKU_28 WH_4 WH_1 19 0
SKU_14 WH_4 WH_5 18 0
SKU_16 WH_2 WH_1 4 0
SKU_25 WH_2 WH_3 8 0
SKU_3 WH_4 WH_3 18 0
SKU_32 WH_5 WH_4 4 0
(base) prishaagarwal@Prishas-MacBook-Pro-2 PythonData % █

```

Python was used to convert inventory shortages into actionable transfer recommendations. For each low-stock SKU–warehouse combination, the system identified another warehouse with sufficient excess stock and calculated the quantity to transfer. In the sample shown, the recommended transfers fully covered the shortage, reducing the remaining supplier replenishment requirement to zero.

```

BEFORE VS AFTER OPTIMIZATION
-----
Total shortage before transfers: 58897
Units resolved through warehouse transfers: 58298
Remaining supplier replenishment required: 599
Percentage of shortage resolved internally: 98.98 %

```

Before inventory rebalancing, the low-stock warehouse positions had a combined shortage of **58,897 units** relative to their reorder points. The Python optimization logic identified that **58,298 units** could be fulfilled by transferring excess inventory from other warehouses, leaving only **599 units** requiring external supplier replenishment. This means **98.98% of the identified shortage could be resolved internally through warehouse rebalancing**, substantially reducing dependence on new supplier orders.

```

SUPPLY CHAIN STRESS TEST
-----
Current risky positions: 3242
Risky positions with 20% demand increase: 6396
Risky positions with 20% inventory reduction: 7296
Risky positions with supplier lead time +3 days: 7823

```

The stress-test simulation shows that the supply chain is highly sensitive to operational disruptions. The baseline contains **3,242 risky positions**. A 20% increase in demand raises this to **6,396**, while a 20% inventory reduction increases it

further to **7,296**. The most severe scenario is a **3-day increase in supplier lead time**, which raises the number of risky positions to **7,823**. This suggests that supplier delays represent the greatest vulnerability among the tested scenarios, reinforcing the importance of lead-time management and proactive inventory rebalancing.

```

PRIORITY & ACTION ENGINE
-----

Priority summary:
Priority
HEALTHY      84209
MEDIUM      3799
HIGH        2965
CRITICAL     277
Name: count, dtype: int64

Recommended action summary:
Recommended_Action
NO ACTION      86463
TRANSFER       4737
REORDER        33
TRANSFER + REORDER  17
Name: count, dtype: int64

```

The Python decision engine classified inventory situations by urgency and recommended an operational response for each case. Most observations were healthy and required no action, while **277 cases were classified as critical** and **2,965 as high priority**. The system identified **4,737 cases where inter-warehouse transfer was sufficient**, while only a small number required direct supplier replenishment or a combination of transfer and reorder. This demonstrates how the model can convert stockout-risk information into prioritized, actionable supply-chain decisions.

TOP CRITICAL INVENTORY CASES							
SKU_ID	Warehouse_ID	Shortage_To_Reorder_Point	Suggested_Donor_Warehouse	Suggested_Transfer_Qty	Minimum_Replenishment_After_Transfer	Recommended_Action	
SKU_29	WH_2	49	WH_5	49	0	TRANSFER	
SKU_3	WH_3	43	WH_4	43	0	TRANSFER	
SKU_41	WH_4	43	WH_2	43	0	TRANSFER	
SKU_43	WH_1	42	WH_4	33	9	TRANSFER + REORDER	
SKU_40	WH_2	42	WH_4	42	0	TRANSFER	
SKU_10	WH_2	41	WH_4	41	0	TRANSFER	
SKU_23	WH_2	40	WH_3	40	0	TRANSFER	
SKU_23	WH_4	40	WH_5	40	0	TRANSFER	
SKU_10	WH_2	40	WH_4	40	0	TRANSFER	
SKU_4	WH_4	39	WH_5	39	0	TRANSFER	

The priority engine ranks the most urgent inventory cases and converts each shortage into a specific operational recommendation. Most critical cases can be resolved entirely through inter-warehouse transfers, while cases with insufficient donor inventory are automatically assigned a combined **Transfer + Reorder** action. This enables management to focus on the highest-priority SKU–warehouse combinations and take a clearly defined response.

NETWORK HEALTH SCORE

Network Health Score BEFORE optimization: 76.61 / 100
Network Health Score AFTER internal rebalancing: 98.03 / 100
Improvement: 21.42 points

The custom Network Health Score increased from **76.61 to 98.03** after applying the internal warehouse rebalancing recommendations, representing an improvement of **21.42 points**. This suggests that the proposed transfer strategy substantially improves overall inventory health by reducing stockout exposure, low-inventory positions, and unresolved shortages. Since this score is a project-specific management KPI rather than an industry-standard metric, its calculation methodology is clearly defined and should be interpreted as a comparative indicator of network improvement rather than an absolute benchmark.

FINANCIAL IMPACT OF INVENTORY REBALANCING

Total shortage value before rebalancing: 720065.34
Inventory value covered through internal transfers: 712740.33
Remaining value requiring supplier replenishment: 7325.01
Percentage of shortage value covered internally: 98.98 %

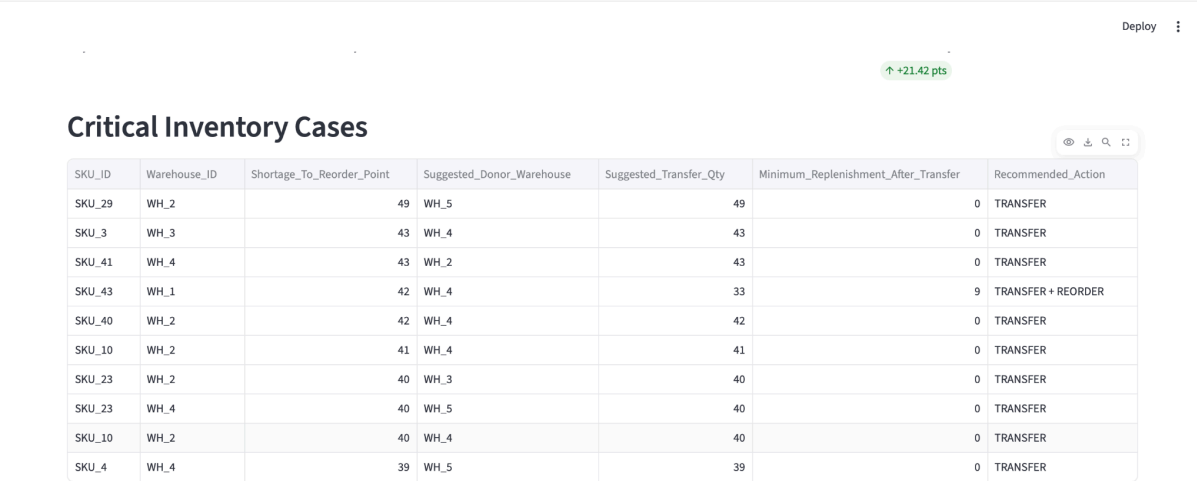
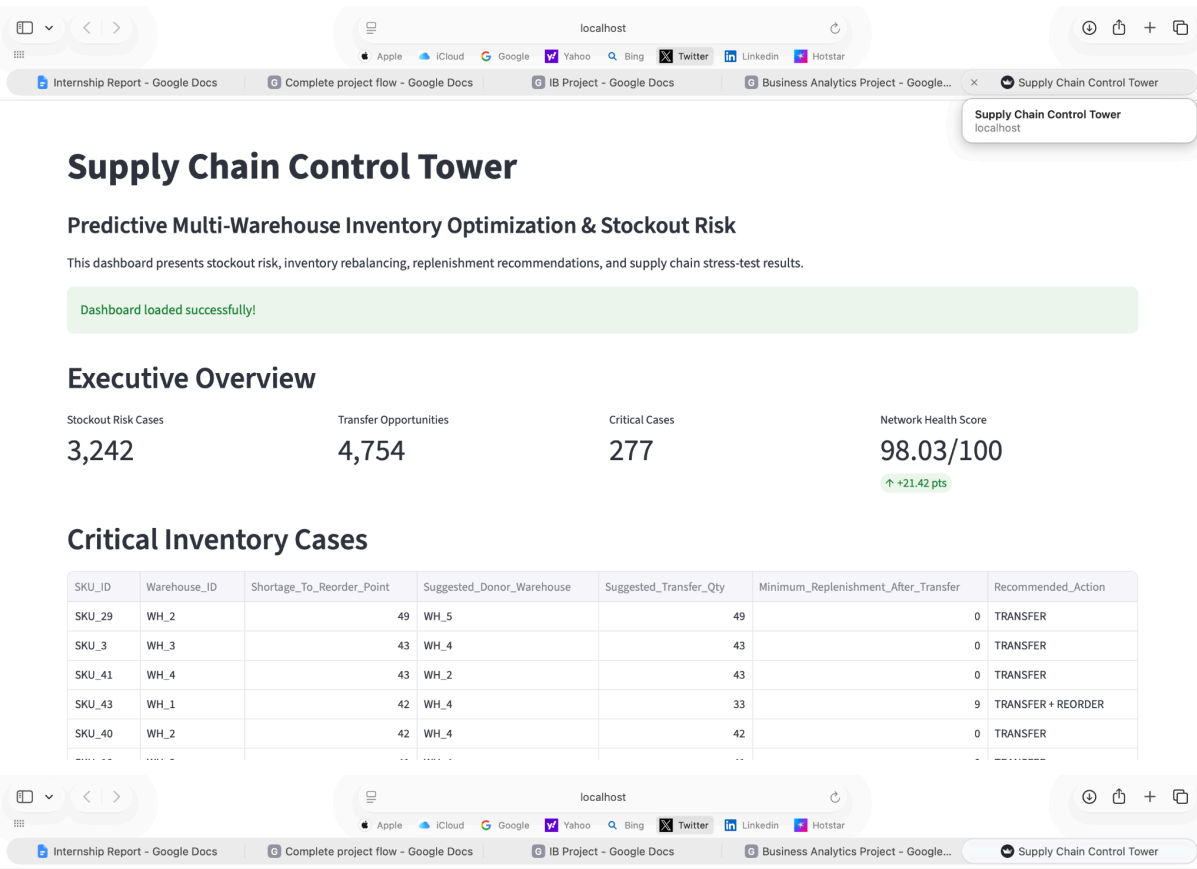
The financial-impact analysis shows that the identified shortage positions represented a total inventory value of **720,065.34** before rebalancing. By reallocating existing excess inventory across warehouses, the system identified **712,740.33** of shortage value that could potentially be covered internally, leaving only **7,325.01** requiring supplier replenishment. This means **98.98% of the shortage value could potentially be addressed through internal inventory rebalancing**. The result highlights the financial importance of coordinating inventory across warehouses before placing new external replenishment orders.

Stage 4: Decision-Support Dashboard Using Streamlit

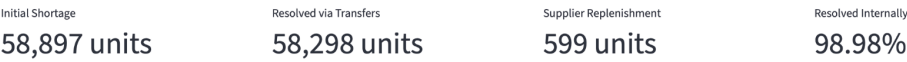
Purpose of using Streamlit:

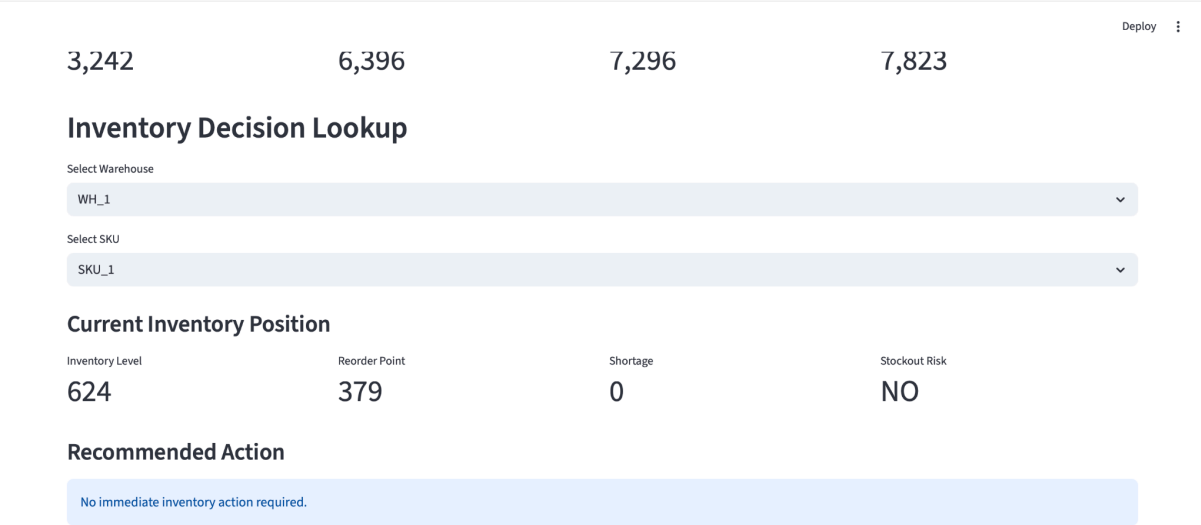
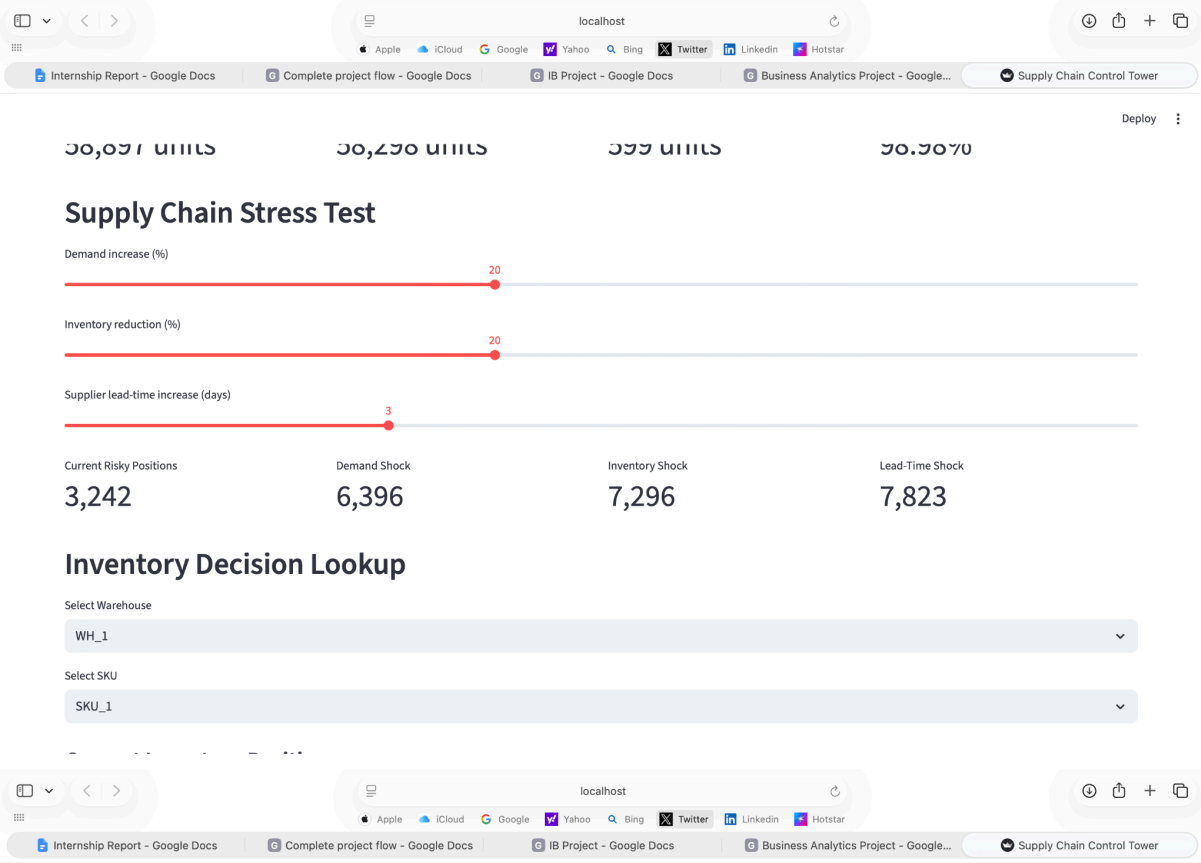
Streamlit is used to convert the analytical results into an interactive management interface. It does not replace the statistical or prescriptive analysis; instead, it allows decision-makers to access the outputs in a simple operational format.

StreamLit Dashboard



Inventory Rebalancing Impact





Key Findings

- Stockout risk was found to be driven mainly by **operational conditions**, especially inventory level, supplier lead time and recent demand. This showed that the problem was not simply that certain warehouses or suppliers were “bad,” but that underlying inventory and replenishment conditions were creating the risk. Business Analytics Project-2
- The predictive stage successfully identified stockout-risk cases, with Logistic Regression achieving **AUC = 0.996** and detecting about **80% of actual stockouts**. This allowed the project to move from only understanding stockouts to identifying where intervention may be required. Business Analytics Project-2
- The Python decision engine then converted these risks into actions by identifying **donor warehouses with excess inventory** and recommending internal transfers before placing supplier orders. As a result, **58,298 of 58,897 shortage units** could potentially be covered through warehouse rebalancing, leaving only **599 units** for supplier replenishment. Business Analytics Project-2
- The decision engine also prioritized inventory cases and recommended whether each position required **Transfer, Reorder, Transfer + Reorder, or No Action**. This made the analysis operationally useful by showing management exactly what action to take for high-risk SKU–warehouse combinations. Business Analytics Project-2
- After applying the internal rebalancing recommendations, the custom **Network Health Score improved from 76.61 to 98.03**, an increase of **21.42 points**. This indicates that the proposed actions reduced stockout exposure, low-inventory positions and unresolved shortages across the network. Business Analytics Project-2
- The financial analysis showed that internal rebalancing could potentially cover **712,740.33 of 720,065.34** in shortage inventory value, leaving only **7,325.01** requiring supplier replenishment. This demonstrates the business value of using existing inventory more effectively before purchasing additional stock. Business Analytics Project-2
- The stress-test analysis showed that the system is especially vulnerable to supplier delays. A **3-day increase in lead time** increased risky positions from **3,242 to 7,823**, making it the most severe scenario tested. This suggests that supplier lead-time management should be a major priority alongside inventory rebalancing.

Limitations

- The dataset represents a modeled/simulated supply-chain environment and may not capture every real-world operational constraint.
- Transfer recommendations do not include transportation, handling or warehouse-transfer costs.
- The rebalancing results represent potential inventory redistribution rather than a fully capacity-constrained mathematical optimization model.
- The Network Health Score is a **project-specific comparative KPI**, not an industry-standard benchmark.
- Predictive performance should be validated on new operational data before real-world implementation.

Conclusion

This project developed an end-to-end **multi-warehouse stockout risk and inventory decision-support framework** by combining diagnostic, predictive, prescriptive and dashboard-based analytics.

The analysis first identified the main drivers of stockout risk, showing that **inventory level, supplier lead time and recent demand** were more important than warehouse or supplier identity alone. The predictive stage then confirmed that Logistic Regression could distinguish stockout and non-stockout cases very effectively, with an **AUC of 0.996** and about **80% stockout recall**. Business Analytics Project-2 Business Analytics Project-2

The project then moved beyond prediction by using Python to recommend specific operational actions. Internal warehouse rebalancing could potentially cover **58,298 of 58,897 shortage units**, leaving only **599 units** requiring supplier replenishment. The custom Network Health Score improved from **76.61 to 98.03**, showing a substantial improvement in overall inventory condition after applying the recommended transfers. Business Analytics Project-2 Business Analytics Project-2

The financial analysis further showed that **712,740.33 out of 720,065.34** in shortage inventory value could potentially be covered internally, while the stress-test analysis identified supplier lead-time delays as the most serious vulnerability among the tested scenarios. Business Analytics Project-2 Business Analytics Project-2

Overall, the project demonstrates that a supply chain can improve inventory decisions not only by predicting where stockouts may occur, but also by **reallocating available inventory, prioritizing critical cases and recommending the most appropriate response before placing additional supplier orders**. The Streamlit control tower brings these outputs together into an interactive format that can support faster and more informed inventory decisions.